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TOPICAL REVIEW

Comprehensive Review of Collaborative Data Caching in Edge Computing

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ABSTRACT Edge computing has gained significant attention due to the swift development of wireless communication technology. As more smart, data-intensive applications and Internet of Things (IoT) devices are used, the amount of data traffic grows at an exponential rate. This means that we need to manage data effectively so that services can be scalable, bandwidth-efficient, and latency-sensitive. Collaborative data caching has emerged as an essential technique in this context to meet the storage and retrieval needs of edge computing systems. We thoroughly examine the current collaborative caching techniques in edge computing, with a focus on recent developments and emerging approaches. We have categorized collaborative caching strategies into four main types: models based on stochastic, game theoretic, and mathematical methods to deal with uncertainty in network conditions and optimize resource allocation; machine learning-based models that employ artificial intelligence (AI) for content popularity prediction and cache optimization; heuristic models providing lightweight solutions for cache placement and replacement; and hybrid models combining multiple strategies. Each of these models is intended to maximize performance under various network and data conditions. We conclude by looking into some of the open problems and difficulties with edge caching to promote further research in this field.

INDEX TERMS Cache consistency, cache optimization, collaborative caching models, edge computing, IoT, machine learning.

I. INTRODUCTION

Technological advancement has a profound influence on various aspects of daily life. Several domains of human life, like communication, industries, education, hospitals, media, business, banking, smart vehicles, etc., were largely influenced by technology [1]. The broad acceptance of IoT has resulted in a substantial rise in the number of heterogeneous smart devices connected through the Internet, enabling numerous applications across various domains and generating large amounts of data [2]. The enormous magnitude of data increases processing delays and contributes to significant network traffic. Gartner, a research firm, forecasts that by 2025, over 75% of data will be produced and modified outside the data centers or cloud environments [3]. As per Ericsson, the increase in mobile data use is anticipated to reach 76% with a global data volume of 563 Exabyte per month by the year 2029 and a single mobile device in India alone is projected to generate almost 75 GB of data [4]. Managing, processing, and analyzing this data presents major difficulties particularly as IoT applications usually call for quick data distribution and secure, private data handling at the network edge.

Edge computing has transformed into a technology that works with cloud computing to address these issues. By moving data processing and cache capability to the network edge, near to end users, edge computing effectively handles network congestion, lowers latency, increases bandwidth efficiency, and improves user quality of service (QoS) [5], [6]. This paradigm change is significant for IoT applications that need to distribute data quickly or that handle private information locally. Edge computing uses caching to reduce latency and response time, improving the user's quality of experience (QoE) [7], [8]. Given the large volume of data generated and transmitted across edge nodes throughout the network, the need for data caching in edge computing becomes

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apparent. When it comes to caching data in edge computing, a variety of elements are taken into consideration, such as data popularity, caching costs, computing power, energy usage, quality of service, etc. [9], [10], [11]. The location and mobility of users on Micro Base Stations (MBSs), Small Base Stations (SBSs), or end devices can also be taken into account while performing edge caching [12], [13]. A STAR-RIS and SIM enabled natural actor-critic agent is utilized to maximize the data rate of the system while adhering to base station power restrictions and QoS requirements [14]. Various cache invalidation and replacement strategies are used to refresh the cache's contents at regular intervals [15], [16]. Caching can be implemented in edge computing using many methods. Each possesses its advantages and disadvantages. Collaborative data caching in edge computing has garnered significant interest among researchers aimed at optimizing data availability and retrieval rates while reducing the burden on the main network. Collaborative data caching in edge computing involves several edge nodes cooperating to effectively store, share, and manage data. This method lowers the frequency of data retrievals from distant cloud servers and provides quicker access to frequently requested material. Collaborative caching improves system scalability and resource utilization, speeds up end-user response times, and minimizes network congestion by distributing cached data throughout the network.

Nevertheless, implementing collaborative data caching in edge computing systems has specific challenges. Key issues to address include managing cache replacement policies, efficiently allocating limited cache resources, ensuring data consistency across distributed caches, and complying with security and privacy requirements. Moreover, caching tactics are complicated by the dynamic characteristics of IoT-generated data and the diverse nature of IoT devices, necessitating innovative solutions and robust frameworks. The aim of this review article is to give an in-depth overview of the current state of edge computing research on collaborative data caching. It looks at the many approaches, frameworks, and algorithms suggested to improve cache performance and handle the specific challenges that the IoT ecosystem brings about. The report also investigates new trends such including artificial intelligence (AI) and machine learning (ML) into caching systems and stresses prospective future research area. This study aims to provide insightful analysis by synthesizing the current body of knowledge for academics and practitioners trying to build more efficient, scalable, and safe collaborative caching systems in the developing environment of IoT-based edge computing. Figure 1 shows the taxonomy of this review paper. The primary contributions of this review are as follows:

- Presenting an in-depth review of existing collaborative caching strategies in edge computing.
- Structuring the collaborative edge caching techniques into four different groups: model-based, deep learning-based, heuristic-based, and hybrid approaches.
- Examining critical aspects and performance metrics of collaborative content caching to enhance future caching strategies.
- Presenting future directions and challenges in collaborative caching within edge computing.

The following is the article's remaining section: Motivation and background of the study are discussed in Section II, Section III gives a summary of related review work, Section IV provides the methodology of work, Section V describes the metrics employed to evaluate the performance of collaborative data caching, Section VI presents various collaborative caching models, Section VII outlines the potential open issues and future directions in collaborative caching and Section VIII provides the conclusion of the work.

II. BACKGROUND

A. FUNDAMENTALS OF EDGE COMPUTING AND IOT

The emergence of the IoT signifies a substantial transformation in device interaction and communication. IoT facilitates the interconnection of a diverse range of physical items, from basic sensors to intricate machinery, enabling effortless data interchange via internet. This connectivity supports the creation of a smart environment, including smart homes, cities, and businesses where devices may operate independently and collaboratively to improve efficiency, safety, and quality of life [17], [18]. However, data creation has surged dramatically owing to the extensive use of IoT devices. This data explosion significantly undermines conventional cloud computing techniques, necessitating real-time processing and low-latency communication [19]. Due to inherent latency and bandwidth constraints associated with transmitting large data volumes to and from remote servers, cloud computing often falls short of fulfilling the rigorous requirements of contemporary IoT applications reliant on centralized data centers. Edge computing has emerged as a transformative solution by facilitating data processing near its source at the network edge, in proximity to IoT devices. Edge computing effectively reduces latency, enhances response times, and alleviates the network's bandwidth requirements by decentralizing data processing and positioning computational resources near data generation sources.

B. ROLE OF EDGE COMPUTING IN IOT

In the IoT ecosystem, traditionally using the centralized cloud-centric architectures, edge computing plays an important part as it addresses key challenges of latency, bandwidth constraints, and privacy issues resulting from the generation and sharing of huge volumes of data and its processing by these centralized cloud servers. Edge computing allows data processing at the edge of a network closer to the source and tackles these issues. In time-sensitive applications with limited bandwidth, the edge computing technique improves efficiency by reducing latency, optimizing bandwidth through local data filtering, and strengthening security and privacy by limiting data exposure. Furthermore, minimizing data

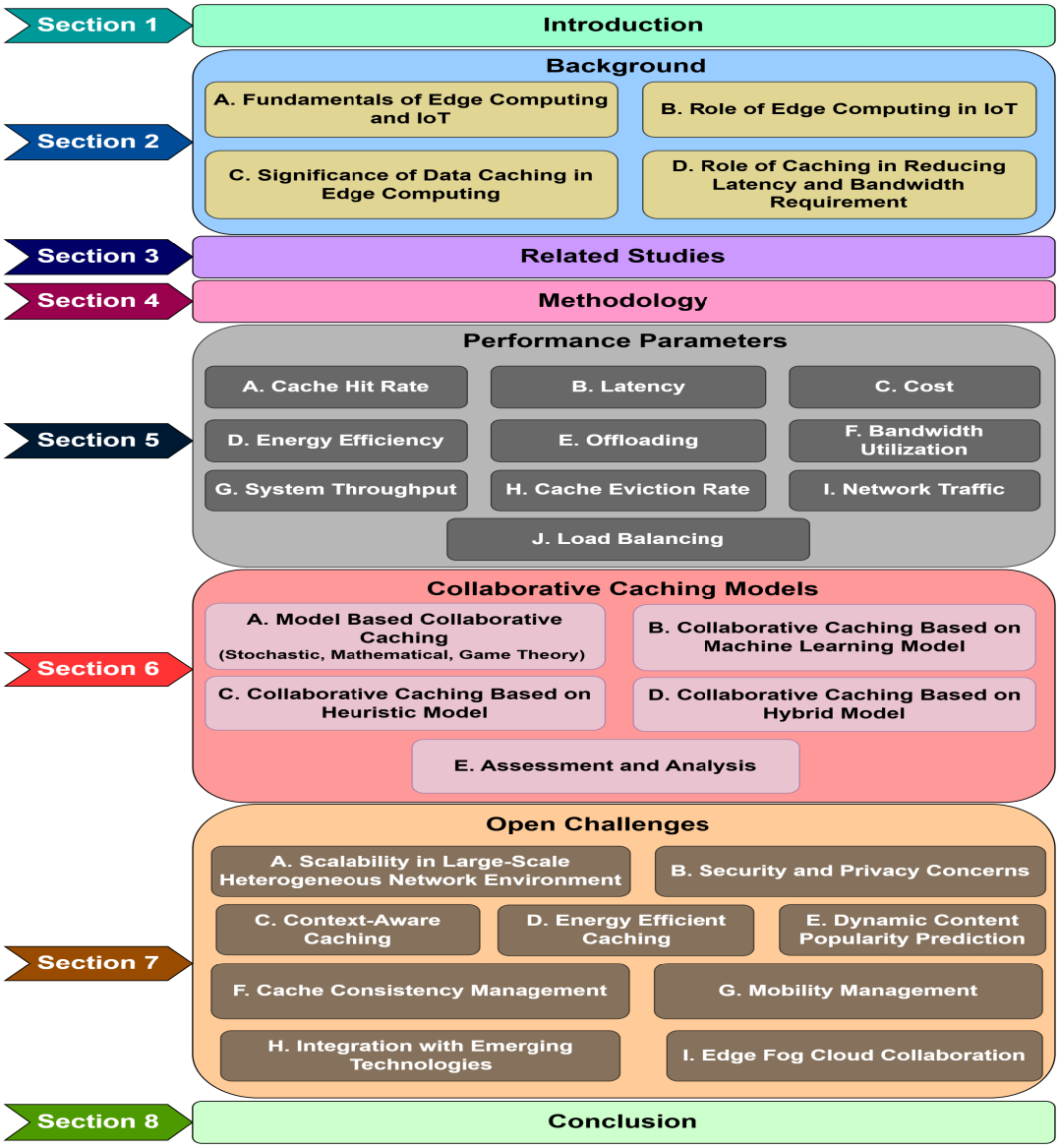


FIGURE 1. Taxonomy of paper.

exposure and facilitating scalable IoT deployments via distributed resources improves security and privacy. By enabling continuous operation even in case of cloud connectivity failures, edge computing increases the need for real-time processing to facilitate prompt decision-making and enhance durability to network attacks. These benefits demonstrate the significance of edge computing in IoT applications and other domains like driverless cars and smart cities, making it an essential technology for the expanding IoT landscape.

C. SIGNIFICANCE OF DATA CACHING IN EDGE COMPUTING

In traditional cloud computing models, data travels long distances to centralized data centers. However, in edge computing, data caching allows storing the frequently accessed content at the edge, near the devices or end users, thus optimizing efficiency and performance, primarily within the IoT landscape [20]. Data caching lowers the need to fetch repeatedly requested content from far-end servers. In delay-sensitive applications such as online gaming, video streaming, autonomous vehicles, and industrial automation, this technique remarkably reduces latency, resulting in a noticeable improvement. Therefore, by minimizing transit time, edge caching facilitates near-instantaneous response, speeds up data access and processing, and enhances system performance in general.

D. ROLE OF CACHING IN REDUCING LATENCY AND BANDWIDTH REQUIREMENT

By minimizing the volume of content transferred between end devices and centralized servers, caching also helps to maximize bandwidth use besides latency reduction. Frequent

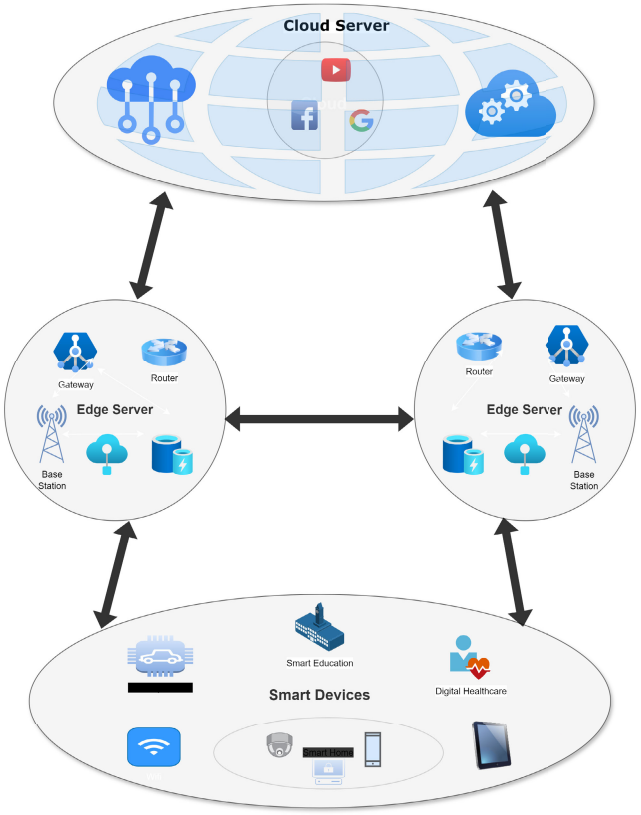


FIGURE 2. Architecture of collaborative data caching in edge computing.

data transfer in traditional cloud architectures requires significant bandwidth, which may be expensive, particularly in the case of limited bandwidth. Edge caching lessens this by locally storing often requested or essential data, therefore relieving network load, preserving bandwidth, and lowering operational costs [21]. Edge computing allows several caching techniques to be used to improve efficiency and speed. Based on their approach to data insertion, management, and retrieval, these techniques may be generally classified as proactive, reactive, centralized, distributed, and cooperative caching, etc. Many complex cooperative caching systems have been developed, including artificial intelligence capabilities with edge computing to improve caching performance, lower latency, and enable AI model deployment [22]. Figure 2 illustrates the architecture of collaborative data caching in edge computing.

III. RELATED STUDIES

Several researchers have conducted detailed reviews of content caching and edge computing. Their reviews on edge computing and its architecture emphasize a comprehensive analysis of edge computing definitions, related concepts, and key applications. Additionally, researchers have highlighted several aspects related to caching functionalities and challenges involved in edge computing for the design and implementation of data caching, as shown in Table 1.

Yao et al. [7] examine mobile edge caching (MEC) techniques to improve user experience in fifth-generation (5G) networks while reducing network traffic. It analyzes essential elements such as the locations of caching, the criterion for caching, and contrasts various caching methodologies. The caching job process is segmented into stages: content request, investigation, delivery, and content update. The study also delineates obstacles and proposes prospective research avenues to meet the increasing demands on mobile networks.

A review in [8] delineates the recent advancements in content caching in MEC, addressing several concerns pertinent to data caching. This encompasses cache location, caching insertion and eviction policies, compute offloading, caching system behavior, and caching optimization in wireless networks. The author identifies many critical concerns about content caching in MEC.

Cao et al. [23] provide a comprehensive overview of a three-tier architecture for edge computing. The article addresses several aspects pertaining to edge computing, encompassing architecture, compute offloading, mobility management, resource allocation, security, privacy, authentication, and access control. Additionally, it offers several application possibilities across diverse domains in edge computing.

Wu et al. [24] presented a review of the overall edge computing caching process, including optimization of caching location, caching policy design, and the delivery of caching material. They argued cache location with users, service providers, and network operators to optimize resource use and edge node usage. They examined several distribution systems and caching methods while discussing challenges and future prospects.

In [25], researchers examined proactive caching methods in edge computing. They first examine several parameters of content caching, including energy and spectrum efficiency, cache hit probability, throughput, latency, and cost. To achieve this, focus on several kinds of proactive caching techniques that may be integrated into IoT edge settings to enhance system performance. Furthermore, it provides insights into the prevailing challenges and issues, along with the applications of proactive content caching in edge computing.

Reiss-Mirzaei et al. [26] provided a comprehensive review of socially responsible edge content caching strategies. The research examines socially conscious edge caching strategies that take into account user behavioral traits. It also includes a discourse on many methodologies, including those grounded in game theory, machine learning, modeling, and heuristic approaches.

Emphasizing cache systems and their advantages for improving user data delivery, authors in [27] provide a detailed evaluation of MEC. It assesses many MEC performance criteria while looking into optimization initiatives to raise system performance. Moreover, three categories define cache modeling techniques: machine learning-based, game-theoretic, and information-theoretic ones. Additionally included are prime application examples and open problems, which inspire the

next research to maximize MEC in dynamic network settings.

Li et al. [28] conducted a survey of advanced content caching strategies, emphasizing critical elements such as caching criteria, selection methodologies, and approaches. They categorize these schemes into specific classifications, presenting a taxonomy, and thereafter conducting a qualitative comparison to emphasize essential qualities and characteristics. The report also outlines obstacles in practical implementation, open research issues, and areas necessitating more investigation to enhance current content caching technologies.

Jazaeri et al. [29] thoroughly examined the caching techniques using software-defined networking (SDN) in the context of IoT-based edge computing. They present a comprehensive classification of caching techniques, highlighting their characteristics and assessing the performance effectiveness of different edge IoT and SDN-based caching strategies. The discussion encompasses methods for content selection, cache insertion, cache placement, compute offloading, performance metrics for caching, and the application of SDN in edge caching, as well as challenges and future research directions.

Based on many criteria, authors in [30] provide the thorough classification of caching systems. They examine innovative content caching techniques developed particularly for MEC environments by closely analyzing the key components of present systems. They also find and discuss outstanding research issues and problems that need deeper investigation in order to enhance content caching strategies.

The discussion in many articles lacks focus on recent advancements in Mobile Edge Computing (MEC). Many reviews focus primarily on theoretical aspects, neglecting practical challenges and real-world implementation issues. While some papers cover a broad range of key edge computing techniques, they often omit explicit discussion of collaborative caching mechanisms. Although several studies address a wide array of key edge computing strategies, they frequently neglect to explicitly discuss collaborative caching mechanisms. Furthermore, they may not integrate the most recent developments about the subject at hand. These studies, however extensive in their overview, fail to deliver detailed evaluations of the impact of collaborative data caching. The lack of focus on the area of interest and insufficient cross-comparison and empirical evaluation restrict the capacity to draw definitive conclusions.

IV. METHODOLOGY

This section outlines the methodology employed for the collecting, assessment, and synthesis of pertinent research on collaborative data caching in edge computing. Prominent search terms encompassed "collaborative caching," "Edge Data Caching," "edge computing," "IoT," "IoT Edge," and "machine learning," while significant digital repositories such as IEEE Xplore, SpringerLink, ACM, and ScienceDirect were explored. The inclusion criteria emphasized peer-reviewed works published from 2021 onward concerning caching methods, performance measures, and future technologies in edge computing. Studies were eliminated from the analysis if they addressed caching circumstances irrelevant to the current research or if they were beyond the scope of collaborative edge caching. The chosen literature was subsequently classified according to the predominant methodological technique utilized. These categories include models based on stochastic, game theory, heuristic algorithms, deep learning approaches, and hybrid methodologies that integrate various techniques, with major insights outlined, focusing on obstacles, performance enhancements, and future research objectives.

V. PERFORMANCE PARAMETERS

The stream of data caching in edge computing has been an area of great interest and has become a focal point of research due to the rapid growth of information technology in various industries. With this growth, a substantial volume of data has been created and shared through networks, resulting in a surge in traffic across the access network. Addressing this requires more bandwidth and other resources to be deployed in the network. To this end, caching in edge computing is viewed as a solution to many of these problems. This is particularly true if we utilize the caching capability of each edge server collaboratively; performance can be improved to a much higher level. The following section presents several parameters that can be adopted to optimize the performance of collaborative caching.

A. CACHE HIT RATE

The effectiveness of cache systems varies in management and in improving their use concerning the available cache spaces. The cache hit rate (CHR) shows how efficiently the cache is serving requests and subsequently minimizing delay and improving overall performance. A higher CHR leads to better cache utilization and a more effective caching strategy. Cache size greatly affects cache hit ratio; the larger the cache, the greater the cache hit ratio. Larger cache sizes incur more cost. Hence, the optimum cache size should house commonly retrieved data. Generally, to increase the cache hit rate, more popular and frequently accessed data is stored, and appropriate cache replacement algorithms are employed to evict less frequently accessed content.

Recent developments in edge computing show notable breakthroughs in CHR optimization using several technological approaches. Modern research clearly shows a shift from conventional caching techniques to complex AI-driven solutions. In dynamic vehicle contexts [31], a hybrid technique combining vehicle clustering with LSTM-based prediction and Q-learning showed considerable gains in cache hit rates. A DRL-based cooperative distributed system [32] addressed the issue of microservice caching in 5G-enabled IoT networks, integration of multiple AI technologies with collaborative filtering [33] and K-order Markov Chain

TABLE 1. Summary of related review work.

Ref	Year	Main Focus	Limitation	Focus of Study
[7]	2019	A comprehensive MEC review targeting reduced mobile core and backhaul load due to rising traffic, analyzing caching locations, criteria, and phases, comparing schemes, and proposing future research to enhance mobile network efficiency.	The article lacks coverage of collaborative caching techniques, essential for enhancing efficiency across multiple edge nodes.	Mobile edge caching
[8]	2020	Current trends in MEC and content caching with a focus on cache insertion and replacement rules, evaluate the effects of edge, fog, and cloud computing on latency, and pinpoint important issues and potential avenues for future MEC research.	The survey does not delve deeply into specific real-world deployment challenges of MEC and content caching technologies and lacks validation of the discussed approaches.	MEC and content caching
[23]	2020	A thorough analysis of edge computing, covering its definition, architecture, key technologies, security, and privacy, while comparing it to cloud computing. It talks about the benefits and looks at many uses, including gaming, smart homes, transportation, industrial IoT, and content delivery.	It gives a broad overview of edge computing, but it doesn't go into detail on caching or how related concepts are implemented; instead, work is limited to edge computing architecture.	Overview of Edge Computing
[24]	2021	A thorough analysis of edge caching to enhance user experience and network efficiency, cover placement optimization, policy creation, and the content delivery process. It looks at the stages of material delivery, identifies current issues, and makes recommendations for future lines of inquiry.	The study does not discuss the aspects of collaborative caching and its advantages in edge computing.	Total process of edge caching
[25]	2021	A systematic review of proactive content caching strategies in edge computing, detailing key approaches, performance metrics, techniques, and assessment tools, while identifying open challenges and future research directions for enhancing proactive caching efficiency.	The study does not deeply examine collaborative edge caching and integrating emerging technologies like 5G or IoT with proactive caching techniques.	Proactive content caching in edge computing
[26]	2023	Social-aware caching techniques in MEC are reviewed systematically by grouping them into several categories. Important facets of socially aware caching were highlighted for further study.	Limited exploration of privacy and security concerns related to socially conscious edge caching in dynamic and large-scale networks.	Social-aware perspective in edge caching
[27]	2023	A systematic review of caching strategies, caching locations, highlighting various performance metrics, classification of caching models and Identification of open challenges for future research.	The study does not sufficiently address the complexities and challenges associated with collaborative caching in edge environments.	Mobile edge caching
[28]	2023	Introduced a comprehensive and systematic survey of edge caching techniques, focusing on key issues such as caching locations, categorization of caching objects into data, service, and content caches, while addressing open challenges and future directions in edge caching.	Minimal exploration of dynamic and collaborative caching across multiple edge nodes, as well as its real-time adaptation to rapidly changing network conditions.	Key issues and challenges of Edge Caching
[29]	2024	The review outlines caching strategies across SDN-IoT architecture, categorizing approaches, comparing benefits, and highlighting challenges, solutions, and research directions in EC-SDN-IoT networks.	It lacks a detailed exploration of collaborative caching mechanisms and coordination challenges in these environments.	Caching in edge computing over SDN-IoT
[30]	2024	A comprehensive analysis of current content strategies investigating the critical aspects of existing techniques based on some aspects, such as location, objective functions, criteria, caching methods, supporting algorithms, open issues and challenges.	Inadequate analysis of adaptive caching strategies that dynamically adapt to current circumstances, and challenges in addressing privacy and security.	Content caching in MEC

(KoMC) and LSTM [34] has proven effective in further development of this.

Notably, use of K-order Markov Chain with LSTM for IIoT devices [34] produced considerable improvements in CHR and retrieval latency. Creative combination of RNN, MDP, and FL for coordinated caching for vehicular edge networks [35] and distributed edge computing systems [36] respectively shows significant progress as it shows improved CHR while lowering access latency. These advancements point clearly toward multi-technological solutions that not only improve CHR but also address associated performance parameters, including latency and resource utilization.

B. LATENCY

Collaborative caching techniques aim to minimize the time required by users to access data. Recent advancements in collaborative caching methodologies have shown a considerable improvement in mitigating latency issues, especially in delay-sensitive IoT applications. Examination of contemporary leading solutions indicates a transition from conventional optimization methods to more advanced hybrid techniques. The hierarchical resource management architecture developed in [37] signifies a substantial improvement in latency reduction via improved offloading decisions, especially in complex edge computing settings. This method has been further improved [38] with utility-aware collaborative service caching implementation, which effectively integrates Q-learning to minimize latency and reduce costs. A system employing an Asynchronous Advantage Actor-Critic algorithm integrated with blockchain for the Internet of Vehicles [39] symbolizes a significant advancement in minimizing content delivery latency and augmenting security.

The NDN-based IoT Edge and clustering [40] is remarkable for its substantial decreases in communication latency achieved using innovative data collection and forwarding methods. A cooperative caching mechanism that addresses request routing and content placement [41] demonstrates significant gains in caching performance and content retrieval delay. Recent advancements in [42] and [43], with their heuristic and adaptive methodologies, have pushed the limits, demonstrating substantial enhancements in response times and task completion metrics. While maintaining system efficiency, the integration of edge-cloud collaboration and online caching framework with transcoding capabilities suggested by [44] and [45] shows a notable reduction in latency, resulting in major declines in content delivery time. To minimize latency while preserving system efficiency, several difficult problems continue to exist in optimizing cache placement, controlling network traffic, and guaranteeing timely access to often requested data across scattered edge nodes.

C. COST

Cost in collaborative caching includes various factors, like caching cost, transmission or communication cost, service cost, system cost, and computing cost. Optimization of all these costs is essential to ensure the efficiency of data caching, maintain system performance, minimize energy consumption, and balance the network load. Current developments in cost optimization for collaborative caching systems indicate a notable transition toward multi-dimensional cost management strategies. An examination of modern cutting-edge systems reveals complex strategies that concurrently tackle many cost elements, including storage, transmission, computing, and operational expenditures. The online approach introduced [46] signifies a significant advancement in system cost optimization, especially inside dynamic edge computing contexts that require continual resource allocation adjustments. The study in [47] has further improved this strategy by innovatively integrating relaxation techniques with Lagrange dual decomposition, resulting in significant reductions in task completion costs through optimum resource usage.

Using AI-driven solutions has clearly improved the cost optimization strategies, as shown [48] by means of Q-learning and DQN for application program caching. Their studies revealed significant cost savings for mobile customers while preserving system performance, therefore stressing the efficiency of intelligent caching solutions. Further, the research in [49] has advanced this trend towards AI-enhanced solutions using a hybrid technique that combines LSTM for popularity prediction with Q-learning for dynamic cache optimization, resulting in considerable reductions in search costs while keeping system efficiency. These developments mark a clear change from traditional cost control techniques to more sophisticated, AI-driven systems capable of dynamically balancing several cost factors while guaranteeing best system performance.

D. ENERGY EFFICIENCY

Energy consumption is one of the critical factors in resource-constrained devices such as IoT, mobile devices, and edge servers. It is described as the power that devices and networks consume to store, retrieve, and manage data in an edge or cloud environment. Systematic and effective techniques for energy management should be adopted for longer operations of the devices, reduce the operating costs, and maintain the service quality. Recent advancements in energy-efficient collaborative caching indicate substantial progress in tackling the crucial issue of power usage in resource-limited edge computing settings. Analysis of cutting-edge solutions indicates a transition from conventional energy management methods to advanced, multi-objective optimization procedures. A novel mechanism for processing raw entity state data [50] signifies a substantial advancement, attaining a balanced optimization of energy efficiency, enhancing search precision, and minimizing delays. Their methodology illustrates how contemporary caching techniques can efficiently address the conventional tradeoff between performance and energy usage.

The domain has progressed further with the application of DQN-based algorithms for dynamic IoT node management and workflow scheduling [51]. Their methodology signifies a pivotal advancement in the concurrent optimization of energy efficiency, makespan reduction, and communication cost minimization, especially prominent in intricate edge computing contexts. The merging of on-path and off-path caching techniques [52] exemplifies the forefront of energy-efficient caching solutions, illustrating how hybrid methods can remarkably decrease energy consumption while reducing redundant content transfer. These developments signify a distinct trend toward integrated solutions that enhance energy efficiency while upholding superior performance standards across other essential metrics, indicating a fundamental transformation in the balance between energy consumption and system performance in contemporary edge computing environments.

E. OFFLOADING

Offloading is the process of transferring computational tasks, processing requirements, and storage from the user devices to more capable systems like edge servers or the cloud, and as a result, reducing the burden of the end nodes. End nodes, such as mobile or IoT devices, have limited processing power, storage capacity, and battery life. Offloading efficiently helps conserve available energy and power in resource-constrained nodes. The latest advances in computational offloading methodologies illustrate considerable progress in addressing the performance issues of resource-limited edge computing environments. An examination of contemporary state-of-the-art solutions indicates a transition from conventional offloading strategies to more advanced, integrated methodologies that combine several optimization techniques. The approach in [53] marks a significant progress in the

integration of systematic offloading and managing migrant caching, hence lowering network latency and access times and preserving system efficiency. This approach shows how modern offloading methods might effectively balance compute distribution with cache enhancement.

The domain has advanced further with creative integration of Gradient Descent Allocation with Game Theory [54], therefore enabling a significant improvement in computational offloading choices as well as resource allocation decisions. The trend towards hybrid solutions has advanced significantly with the integrated integer nonlinear programming method [55], which effectively combines Gibbs sampling with deep Q-networks to concurrently improve job offloading and service caching in vehicular edge computing networks. Emphasizing the efficiency of modern offloading techniques in concurrently addressing several performance criteria, their study shows notable improvements in system cost reduction and task execution speed. These developments point clearly toward integrated systems that increase computational distribution while maintaining high performance criteria across key metrics like latency, energy economy, and resource usage.

F. BANDWIDTH UTILIZATION

Bandwidth usage refers to the strategic application of available bandwidth to achieve certain goals. It examines the efficiency of data movement across edge nodes, cloud servers, and user devices utilizing the existing network capacity. Caching solutions are designed to progressively decrease total bandwidth utilization by lowering the necessity to get data from distant sources. By caching popular and often accessed content close to the user at the edge server, transmission of necessary content over long distance is scaled down, optimizing bandwidth usage. In order to improve QoS, lower latency, and speed up data retrieval on bandwidth-constrained mobile and IoT networks, bandwidth optimization involves more than just cache placement and replacement tactics. It also involves procedures to minimize needless data transfer throughout the network.

Research in bandwidth optimization for edge computing environments exhibits considerable success in addressing the difficulties of efficient data transmission and resource use. Analysis of cutting-edge solutions indicates a transition from conventional bandwidth management methods to advanced multi-objective optimization techniques. A resource management strategy for WiFi-cellular heterogeneous networks constitutes a substantial advancement [56], effectively merging job offloading with service caching to concurrently enhance energy efficiency and minimize delay while preserving optimal bandwidth utilization. Their methodology illustrates how contemporary caching methods can proficiently equilibrate various performance parameters in diverse network contexts. A collaborative entity state data caching technique [57] represents a significant progress in the simultaneous optimization of cache placement, computing resource allocation, and bandwidth usage. The trend towards integrated solutions has been advanced with the hierarchical resource management paradigm [37], which merges latest start time-based scheduling with dynamic programming to attain complete optimization across several performance indicators. Their study shows how modern bandwidth optimization strategies may effectively address the complex interaction of offloading decisions, workload scheduling, and caching mechanisms, all while guaranteeing suitable bandwidth consumption. These developments indicate a basic change in the integration of bandwidth optimization with general system performance in modern edge computing environments since they indicate a clear transition towards comprehensive strategies that improve bandwidth usage while preserving enhanced performance benchmarks across other important metrics.

G. SYSTEM THROUGHPUT

Throughput quantifies the rate at which data is transferred over the network. It is defined as the quantity of user requests successfully fulfilled within a specified timeframe. System throughput assesses the efficiency with which the network manages data requests, executes tasks, and distributes content to users. Current advancements in throughput optimization for edge computing indicate substantial progress in enhancing data transmission efficiency and system performance. An examination of contemporary state-of-the-art solutions indicates a transition from conventional throughput management strategies to more advanced procedures that incorporate several performance targets. Especially the development of throughput-optimal control technique [58] using data-intensive distributed compute network control framework. Their approach shows how today's caching systems can efficiently balance throughput optimization with resource use by combining effective database placement algorithms with max throughput database placement.

The aggregation of throughput optimization with additional performance measures signifies a notable advancement in collaborative caching solutions. Recent research trends suggest that effective throughput increase necessitates a comprehensive approach that encompasses data transmission rates, cache management efficiency, content distribution methodologies, and node collaboration. This multi-faceted strategy for throughput optimization indicates that forthcoming advancements in this domain will likely focus on cohesive solutions that sustain high throughput while concurrently tackling other critical performance metrics such as latency reduction and quality of service improvement, signifying a significant shift from traditional single-metric optimization.

H. CACHE EVICTION RATE

Stale data in the cache memory must be cleared periodically to make space to store newer content. The cache eviction rate is the speed at which cache content is cleared to accommodate fresh data. Since the cache is often of limited size, it is

essential to maintain optimal cache size to guarantee the availability of frequently accessed and most essential data. Recent advancements in cache eviction algorithms demonstrate significant progress in balancing content freshness and system performance in edge computing environments. An examination of contemporary state-of-the-art solutions indicates a transition from conventional time-based eviction strategies to more advanced, context-aware methodologies that simultaneously account for many performance measures. A hybrid approach represents a significant breakthrough in this area by successfully fusing popularity-driven eviction strategies with probabilistic caching [59]. Their application demonstrates how modern eviction strategies may optimize many performance metrics at once, leading to greater cache hit rates, reduced latency, and enhanced route stretch ratios.

Cache eviction techniques in MEC have been much improved by the predictive collaborative replacement approach [60]. This method significantly advances the use of predictive analytics with effective replacement algorithms, therefore enhancing CHR and reducing latency. These developments show a clear trend towards integrated solutions that improve the eviction process while maintaining exceptional performance throughout several system criteria. The effectiveness of these approaches suggests that future developments in cache eviction algorithms will most likely highlight predictive and context-aware solutions that may adaptively reconcile content freshness with system performance requirements in dynamic edge computing environments.

I. NETWORK TRAFFIC

In collaborative data caching, minimizing network traffic is often regarded as the highest priority task, as excessive data transfer may increase latency, demand more bandwidth, and cause potential network congestion. To control and reduce the network traffic, appropriate caching strategies can be implemented by storing regularly accessed content closer to users at edge nodes, resulting in lower data flow due to fetching content from the remote server. Recent advancements in network traffic optimization for edge computing environments show substantial progress in tackling the issues of data transfer efficiency and network congestion. Analysis of contemporary state-of-the-art solutions indicates a transition from conventional traffic management strategies to more advanced, AI-driven procedures. The combination of DRL with FL shows a significant development [36] that efficiently reduces communication cost and improves cache hit rates at the same time. Their approach shows how modern traffic management systems could effectively balance several performance objectives in distributed edge computing environments.

A two-stage caching solution for adaptive bit-rate video streaming represents a significant leap in the management of network traffic for multimedia applications [61]. This methodology resulted in considerable decreases in backhaul connection load while enhancing initial buffer delay, illustrating how advanced caching solutions can both optimize network traffic and improve user experience. These advancements reveal a basic change in how traffic management is handled in contemporary edge computing environments since they show a clear trend toward integrated solutions that maximize not only network traffic but also maintain excellent performance across other important criteria. This evolution suggests future advancements most likely centered on multi-objective, AI-enhanced systems for managing network traffic.

J. LOAD BALANCING

Load balancing is the process of splitting data requests and caching activities among several edge nodes or servers to prevent congestion and optimize resource utilization. In an edge computing environment, computing resources and cache storage are distributed across several nodes; hence, to achieve optimal system maintenance, effective load balancing is required. Load balancing solutions for edge computing environments show outstanding development recently in addressing problems of distributed resource management and system optimization. A hierarchical data transmission strategy represents a significant breakthrough, successfully integrating adaptive load balancing with optimization of resource allocation [62], illustrating how load balancing techniques can enhance system performance while ensuring appropriate resource allocation among edge nodes.

Many-objective cooperative caching architecture [63] represents a significant leap in the discipline, facilitating the simultaneous optimization of several performance metrics such as load balancing, prediction accuracy, and network traffic. A coalition game-based load balancing for vehicle edge computing has enhanced the trend towards complete solutions [37] by notably improving simulation performance for C-V2X applications. These developments represent a clear change from simple load distribution techniques to complex, context-sensitive systems capable of adaptively balancing load while keeping optimal performance across several parameters.

VI. COLLABORATIVE CACHING MODELS

Collaborative data caching is an advanced strategy where multiple edge servers or nodes work cooperatively to store, manage, and retrieve content within a distributed network. The collective resources of edge devices are utilized to optimize data delivery, enhance throughput, improve performance, and enhance the network's efficiency in general. By enabling edge nodes to share and exchange cached data, a collaborative caching system removes the need to get data from distant central servers [30]. In the IoT, where vast volumes of data are created and analyzed instantly, this method is quite helpful. This segment divides collaborative caching in edge computing into four categories: model-based, which includes stochastic, mathematical, and game-theoretic methods; heuristic-based; machine learning-based;

and hybrid approaches combining these methods, as shown in figure 3. A tabulated summary of each category is provided at the end of each section. The tables include columns detailing the metrics considered, the network models employed, the algorithms adopted, the datasets and simulations used for implementation, the results obtained, and their limitations.

A. MODEL BASED COLLABORATIVE CACHING (STOCHASTIC, MATHEMATICAL, GAME THEORY)

A stochastic model combines randomness or probabilistic decisions into its process; thus, the output may differ across iterations even with the same starting input because of the random decisions taken during operation. In contexts when several entities, including network nodes or service providers, have control over resources, game theory provides a structure for modeling and evaluating interactions. The reviewed works on collaborative data caching in edge computing, utilizing a model-based approach, exhibit significant improvements across various parameters such as latency, offloading, network efficiency, resource allocation, energy consumption, and user experience. We present some of the referred works on collaborative data caching based on stochastic models here.

The authors have focused on metrics like offloading and caching techniques to minimize latency and access time [37], [53], [54] and cost [46], [47], [54]. To achieve this, a Lyapunov optimization-based online algorithm is exploited in [46], relaxed Lagrange dual decomposition and dynamic programming are adopted in [47], systematic offloading and migrant caching are utilized in [53], and an algorithm based on game theory and gradient descent allocation is taken into consideration in [54], a load balancing algorithm based on coalition game theory [37] and a Max-Throughput Database Placement (MTDP) [58] to significantly enhance QoE, reduce system and access delay, and maximize throughput. The summary of the work on model-based (stochastic, mathematical and game theory) collaborative caching, highlighting the key aspects of each approach, is presented in Table 2.

A few research studies in edge computing environments have concentrated on creating stochastic, game theory, and mathematical models for cooperative data caching. Xia et al. [46] presented an online algorithm that minimizes system costs related to caching, migration, and QoS penalties using Lyapunov optimization to address the collaborative edge data caching problem. Correspondingly, to reduce user cost, Feng et al. [47] concentrated on a two-layer MEC system and suggested a cooperative approach to data caching and computing offloading. To minimize task completion costs and reduce offloading delay, they presented a combination of relaxation, Lagrange dual decomposition (LDD), and dynamic programming (DP) for resource optimization. A framework for intellectual power computing (IPCF) was introduced by Zheng et al. [53] combines systematic offloading and managerial migrant caching (MMC) to reduce network delays and access time while enhancing energy efficiency and resource allocation.

Zhou et al. [54] took a similar approach for resource allocation and offloading computation, employing a collaborative caching mechanism, which optimizes task costs using Gradient Descent Allocation (GDA) and Game Theory (GT) based techniques. The proposed method shows near-optimal performance with lower complexity. These approaches underline important future directions in adaptive resource allocation and multi-server configurations by aiming to enhance the utilization of resources, reduce latency, and improve caching efficiency in edge networks. The study in [37] proposes a hierarchical resource management paradigm for MEC-assisted cellular-vehicle-to-everything (C-V2X) applications, aiming to reduce latency by optimizing offloading decisions, job scheduling, and caching strategies. To enhance simulation performance, the study uses the Coalition Game-based Load Balancing Algorithm (CG-LBA), Dynamic Programming-based Caching Algorithm (DP-CA), and Latest Start Time-based Scheduling Algorithm (LST-SA). Cai et al. [58] presented a method to collaboratively manage communication, caching, and computing resources for data-intensive AgI services in next-generation networks. Complex, multi-pipeline data flows in distributed cloud environments were managed using throughput-optimal control policies called Data-Intensive Distributed Compute-Network Control (DI-DCNC) and efficient database placement and replacement strategies using Max-Throughput Database Placement (MTDP). Future network designs will be able to implement data-intensive AgI applications more successfully because of the suggested solutions' notable increases in service delivery efficiency.

Although these approaches have helped edge cache performance to show notable improvement, several challenges still remain. The dependency on accurate real-time data for system states, as discussed in [46] can be challenging; moreover, the computational complexity might grow with larger network sizes. The method in [47] may not be scalable in dense networks. Research in [37] does not look into collaborative offloading among vehicles and energy consumption optimization. Some studies assume static network conditions and do not fully account for the dynamic and heterogeneous nature of edge environments. While few other studies generally concentrate on individual concerns instead of presenting holistic remedies. Future research should focus on developing adaptable and scalable algorithms that can manage edge networks' dynamic properties. There is an urgent need for more comprehensive evaluations of cooperative data storage systems that take into account security, privacy concerns, and energy efficiency.

B. COLLABORATIVE CACHING BASED ON MACHINE LEARNING MODELS

Three fundamental paradigms in artificial intelligence are machine learning, reinforcement learning (RL), and deep learning (DL). ML is a broad class of algorithms that

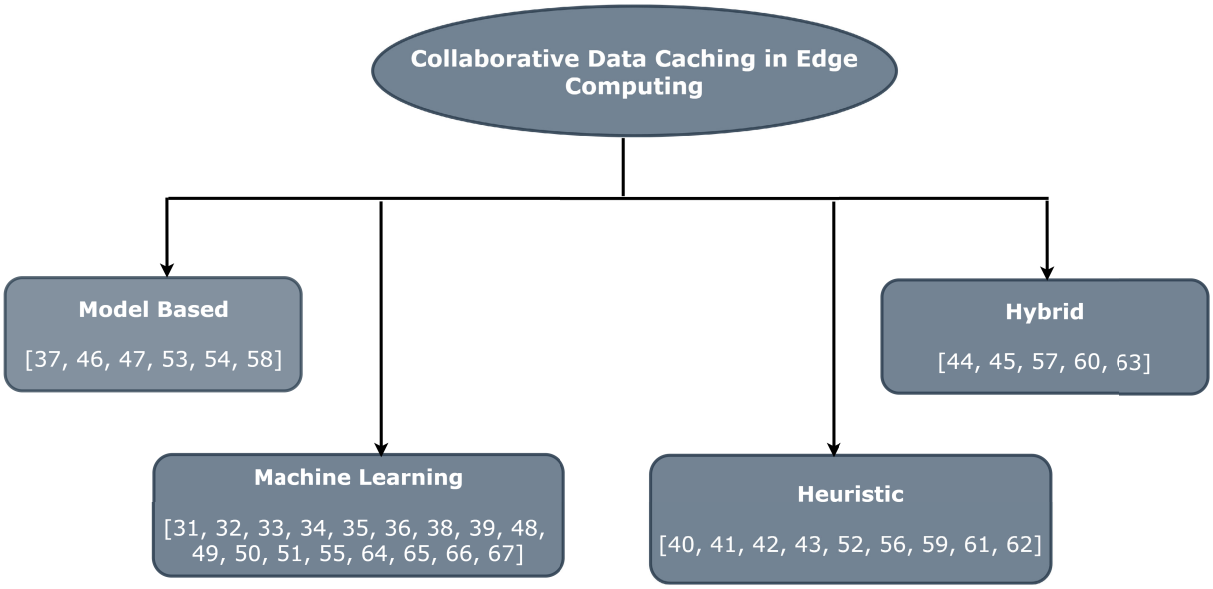


FIGURE 3. Model classification.

use statistical techniques to learn labelled (supervised) and unlabelled (unsupervised) data, allowing systems to use data to identify patterns and make decisions without explicit programming. Deep learning is a subset of ML that automatically learns the hierarchical feature from a large raw dataset using an artificial neural network with multiple layers. DL includes architectures for sequential data like RNNs/LSTMs and CNNs for image processing. Reinforcement learning (RL) is a learning model that facilitates agents to learn optimal policies through trial and error. RL adopts a mechanism based on rewards and penalties to reinforce positive actions while penalizing negative ones, thereby improving the process of decision-making. It is widely employed in domains like gaming, robotics, and autonomous systems. These approaches can be merged, as in deep reinforcement learning (DRL), in which deep neural networks enhance RL's power to learn advanced policies and handle complex state spaces.

These machine-learning models are extensively utilized in collaborative caching to estimate data popularity, optimize cache placement, and improve overall system efficiency. These models can process complex data and offer higher accuracy and greater efficiency in dynamic environments. Some of the referenced works that employ models based on ML, DL, and RL for collaborative data caching are presented here. Several innovative frameworks and techniques have been put forth to deal with the challenging task of collaborative caching in edge computing and IoT systems. Researchers have focused on metrics like cache hit rate [31], [32], [33], [34], [35], [36], [49], [64], and [65], content access latency [31], [32], [33], [34], [35], [39], [50], [55], [64], and [65], task offloading [48], [55], energy overhead [48], [50], [51], QoE [33], [66], content popularity [34], [49], [65], cost [38], [49], [55], data privacy [39], [66], mobility [34], completion time [51], service latency [38], [67], Search accuracy [50] etc to enhance the the performance and effectiveness of collaborative caching technique in edge computing.

Various popular machine learning-based models and techniques have been used by researchers to achieve these goals. For instance, Q-learning is employed by [31], [38], [48], and [49], Deep Q Network (DQN) by [34], [48], and [51], Double DQN and Dueling DQN by [32] and [64], Multi-agent Competitive DQN by [35], Gibbs Sampling + DQN (G+DQN) by [55], Long Short Term Memory (LSTM) [31], [34], [49], Federated Learning (FL) by [35] and [36], Markov Decision Process (MDP) by [32], [35], [39], [55], [64], and [67], Deep Reinforcement Learning (DRL) [36], [39], [64], and [67], K-means clustering by [31] and many other customized algorithm. Table 3 and table 4 summarize the work on collaborative caching based on the machine learning model, detailing the metrics, network models, algorithms, datasets, simulation tools, results, and limitations of the proposed approach.

Wang et al. [31] developed a caching technique for the Internet of Vehicles (IoV) integrating vehicle clustering, LSTM-based content demand prediction, and Q-learning for optimum cache placement, boosting cache hit ratios, and lowering content acquisition delays. Their methodology offers interesting ways to raise QoS in IoV systems by addressing the dynamic character of content popularity in vehicle networks. Tian et al. [32] presented a cooperative decentralized microservice caching system leveraging a DRL-based Distributed Double Dueling Algorithm with Deep Q-Network (D3QN) for independent caching decisions by IoT devices, focusing on MEC-capable IoT devices. The

TABLE 2. Model based Collaborative Caching (stochastic, mathematical, game theory).

Ref	Year	Metrics	Network Model	Algorithms	Dataset Used	Simulation	Result	Limitation
[46]	2021	Caching cost, migration cost, and QoS penalty	MEC	Lyapunov optimization	EUA	Not Specified	Considerable reduction in system costs	Does not consider user mobility
[47]	2021	Total system costs	MEC	LDD, DP	Not Specified	Not Specified	Remarkably reduces total cost	Unrelated to content popularity in heterogeneous networks
[53]	2022	Delay, traffic and access time	Mobile network	IPCF, MMC	Not Specified	Not Specified	Minimize network delays, access time	Challenges in complex offloading and caching due to delay and traffic
[54]	2023	System task costs, communication delay, and traffic	Smart grid	GDA, GT	Not Specified	Not Specified	Enhances cache-hit ratio and reduces content access delays	Edge Collaboration not explored
[37]	2023	Average application completion delay, failure rate resource utilization	C-V2X, MEC	LST-SA,DP-CA,CG-LBA	Not Specified	MATLAB	Significantly improves application completion delay, application failure rate, and resource utilization	Lacks collaborative offloading, does not address energy consumption optimization
[58]	2024	Throughput, routing, processing, and caching for both live and static data	MEC	DI-DCNC, MTDP	Not specified	MATLAB	Optimizes data intensive services through improved packet routing, processing, and replication	Other QoS parameters not extensively considered

approach employs an MDP to model the caching problem and utilizes edge nodes for DRL agent training to reduce computational load on IoT devices. Based on simulation results, the model is successful in terms of lowering fetching latency and increasing cache hit ratio.

Using Q-learning and DQN, Elgendy et al. [48] developed a task offloading and caching model in the multi-user MEC environment to save time and cost of energy. Based on processing capability and job popularity, their solution stores application programs and related code at edge servers, thereby significantly lowering total costs for mobile users. The study of Zhang et al. [64] looked at 5G-enabled large IoT networks for allocating radio resources and collaborative edge caching. The researchers developed novel algorithms based on DRL for Cooperative Content Caching (DRL-CCC) Combined Double DQN, Dueling DQN, and Efficient Radio Resource Allocation centered on the Branch-and-Bound Method (IBBM-RRA) that effectively improved content caching hit ratios and reduced retrieval delays. The performance of the proposed methods is better than baseline algorithms, although scalability in ultra-dense networks remains challenging. Gupta et al. [33] aimed at enhancing user QoE in edge-cloud environments, implementing an AI-enhanced content caching strategy by integrating AI technologies and collaborative filtering with edge, cloud, and information-centric networks (ICN), IoT. Simulations show notable gains with a 12% decrease in retrieval latency, a 28% decrease in average hop count, and a 15% improvement in cache hit ratio. However, a more refined strategy employing larger attributes and varying cache capacities with different Named Data Network (NDN) schemes is considered future work.

For the Cyber-Physical Systems (CPS), a caching strategy addressing Time-Varying Feature Extraction (TVFE) through a hierarchical edge-cloud architecture and collaborative caching based on Entity State data clustering (ESDC) is introduced [50]. This approach processes raw entity state data, utilizing both cloud and edge resources and implementing differentiated data storage, achieving improvements in search accuracy, delay reduction, and energy efficiency. The system, however, needs to focus further on enhancing search efficiency and user privacy protection. An optimal caching policy is formulated in [34] for mobile Industrial IoT (IIoT) devices, using a K-order Markov Chain (KoMC) and LSTM for location and content request prediction. A DQN utilizing reinforcement learning with multiple agents (MARL) is applied to determine the optimal caching policy and maximize rewards. The research demonstrates

improved performance in terms of latency and cache-hit ratio and advises securing training data, employing more exact content popularity prediction via federated learning (FL), and ensuring the privacy of local data between servers. Using Q-learning and an algorithm for Composite Upper Confidence Bounds (CUCB), Huang et al. [38] suggested a collaborative service caching technique deemed utility-aware (UACSC) for IoT applications, displaying UACSC's better performance in optimizing service cost and latency reduction. Still, the study has to go farther into dynamic edge server involvement via incentive systems.

Further innovations by Wu et al. [35] include the Federated DRL-based Multi-agent Strategy of Coordinated Caching for vehicular edge networks. The model employs vehicle-to-vehicle (V2V) communication, recurrent neural networks (RNN) for connectivity prediction, and a multi-head attention mechanism for content popularity modeling. For intelligent caching policy selection, MDP and FL are employed. Experimental results show improved caching hit rates and reduced content access delay by 19.8% and 12.5%, respectively, enhancing data transmission efficiency in dynamic vehicular environments. Chen et al. [67] considered cooperative caching and placement of computation service issues in edge-enabled IoT systems for applications driven by sensing data. Employing an encoder-decoder policy network (EDPN), researchers suggested DRL-based solutions, describing their model as an MDP with limited steps. Two methods are presented: an on-policy reinforcement approach and an off-policy Twin-Delayed Deep Deterministic Policy Gradient (T3DPG) method, incorporating a novel Twin-Q-Delayed with Weight-Averaged (WATQD) algorithm to improve training stability. An MEC and blockchain-based framework for the IoV is developed in [39] to reduce the latency of content delivery and enhance security. The framework employs an Asynchronous Advantage Actor-Critic (A3C) algorithm for smart resource allocation. Simulation results demonstrate its superiority over existing methods. Future research will look into alternate consensus techniques and AI algorithms to enhance performance and scalability in vehicle networks. For customized, secure IoT data services, Wu et al. [66] proposed a cooperative edge-cloud architecture. The system uses a secure k-NN algorithm for encrypted data matching and a personalized rating method with fused time (TFPR) for user-specific services. Reducing service time and enhanced privacy protection are shown via simulations. Future efforts will concentrate on improving tailored safe data services for Internet of Things systems.

Yuan et al. [51] developed an energy-efficient plan for cloud-edge collaboration that utilizes algorithms for dynamically managing IoT nodes and scheduling workflows based on DQN to enhance energy efficiency and minimize makespan and communication costs. Although the scheme performs better than existing methods. However, the model needs to focus on improving AI-based node management and transfer learning integration for intelligent collaboration and scheduling efficiency. Zhang et al. [49] focused on improving cache hit rates and reducing search costs in AI of Things (AIoT) environments by combining static entity popularity prediction with dynamic caching optimization. The method combines LSTM to identify popular things based on user preferences and Q-learning to maximize dynamic caching by minimizing search costs, hence surpassing present benchmark solutions in simulations. Similarly, Li at el. [55] proposed a combined integer nonlinear programming approach for task offloading and service caching to reduce the overall system cost and task execution time for the Vehicular Edge Computing Network (VECN). Gibbs sampling (GS) is combined with a deep Q-network (G+DQN) based on DRL to resolve this complicated optimization issue. Superior performance in minimizing system delay and cost and enhanced resource utility is demonstrated by simulation results compared to traditional approaches, although scalability remains a challenge. Tian et al. [36] introduced an efficient cooperative caching framework (FDDL), which incorporates Federated Learning (FL) to optimize cache hit ratios while reducing communication overhead. A DRL-based cache admission algorithm (DRL-CA) and a lightweight cache eviction algorithm are introduced to manage the cache efficiently while maintaining low training time and communication costs. Emulation shows that FDDL outperforms traditional caching policies and DRL-based algorithms in improving cache efficiency, but the study leaves time-sensitive caching and user mobility for future exploration. The work of Lian et al. [65] explored spatiotemporal graph convolutional models that integrate a Gated Recurrent Unit (GRU) and a Graph Convolutional Neural Network (GCN) to forecast the popularity of content. The method uses GCN to extract spatial information from the MEC server topology and GRU to gather temporal data, hence improving content popularity prediction. Using hierarchical clustering, a simplified method groups MEC servers and maximizes cache placement depending on expected popularity. The system balances prediction accuracy against training time to reduce computational complexity at the cost of some accuracy.

Research on ML-based collaborative caching reveal substantial progress in the optimization of caching, task offloading, and resource management in edge computing environments, particularly in IoV, CPS, IIoT, and MEC. The integration of ML, DRL, and FL enhances network management and cache efficiency and reduces system latency. These techniques enable systems to adapt to the changing network requirements, especially in real-time decision-making scenarios, such as in vehicular networks and smart cities.

The use of DRL in task offloading, caching, and resource allocation has been shown to reduce energy usage and latency and improve cache hit rates. Many studies [31], [32] demonstrate that, in dynamic IoV and IoT networks, DRL can optimize content placement, addressing traffic unpredictability, resource availability, and task popularity. Computing efficiency and user experience have improved

significantly with the use of Q-learning and DQN. Some studies show scalability issues in high-speed vehicles and ultra-dense IoT networks. While these simulation techniques demonstrate significant progress, additional research is necessary to ensure their robustness in large-scale, real-world applications.

Literature examined often revolves around the increasing emphasis on privacy protection, security, and energy economy. Particularly with relation to sensitive user data management, studies conducted in [38] and [66] have recommended approaches to protect data privacy and safe data services across IoT and edge environments. These approaches utilize advanced cryptographic methods like secure k-NN and privacy-preserving federated learning to balance efficiency and security. Privacy measures must be studied more to be practical, especially in IoV and IIoT systems where user mobility and data privacy are crucial.

Numerous studies have investigated predictive models for content popularity prediction and cache placement, including LSTM networks and spatiotemporal GCN. As evidenced in the works of [49] and [65], these models have shown promising results in forecasting content demand and improving cache placement. Combining features with respect to space and time for content prediction is a viable path to raise the accuracy and efficiency of storage techniques in MEC and IoT systems. However, these models are not without limitations. As various studies have shown, the trade-off between prediction accuracy and computing complexity still presents a difficulty that calls for further study.

Though these fields have made great progress, several research gaps remain, notably in the scalability of machine learning-based caching and task offloading. Many models are evaluated in controlled simulation environments, with limited testing in real-world, highly dynamic, heterogeneous network situations. Research should focus on scalable technologies that can manage the complexities of large-scale IoT systems, including seamless edge-cloud interaction. The wide range of user behavior and content demand requires stronger models for predicting cache management, especially in environments with multiple IoT devices and dynamic mobility patterns.

The design of incentive systems for cooperative edge-server involvement is yet another important field of study for next year. As various research, including those in [38] and [67] have shown, the efficacy of cooperative caching and offloading mostly depends on the inclination of edge nodes to participate in resource sharing. The effectivness of large-scale IoT and edge computing systems relies on incentive-based structures that facilitate edge server cooperation while maintaining fairness and providing best performance.

In conclusion, the study indicates notable advancements in caching and resource management for IoT and edge environments; however, considerable challenges persist, especially concerning scalability, privacy, and the resilience of ML models in dynamic real-world scenarios. Future research should prioritize the enhancement of scalability in predictive models, the improvement of privacy-preserving mechanisms, and the design of more efficient collaboration models to advance edge computing applications. These advancements are essential for optimizing performance, minimizing costs, and ensuring the reliability of next-generation IoT and AIoT systems.

C. COLLABORATIVE DATA CACHING BASED ON HEURISTIC MODEL

A heuristic approach uses practical, experience-based, problem-solving techniques to arrive at a good enough solution in a fair amount of time, rather than guaranteeing the optimal solution. Heuristic methods are often employed when exact methods are computationally expensive or impossible for large-scale problems. These models provide a mechanism to practically solve complex problems by employing a methodology based on trial and error, rules of thumb, or informed assumptions. Heuristic models are widely used in collaborative data caching due to their simplicity and efficiency in real-time decision-making, such as cache placement and content retrieval. Here, we look through some of the mentioned research on heuristic-model-based collaborative content caching.

The referred articles in heuristic model-based caching have considered several commonly explored metrics for collaborative content caching. A few of the most often investigated metrics that significantly affect caching performance include latency [40], [41], [56], [59], [61], response time [42], [62], QoE [52], [61], cache hit ratio [59], [61], cost [40], [62], energy consumption [52], [56], etc., to improve the overall system performance. To minimize communication latency, content access delay, and cost, authors have explored Forwarding Interest Base (FIB) establishment [40], Primal-Dual Algorithm (PDA) and a Greedy Approximation Algorithm (GAA) [41], hierarchical data transmission strategy [62], hybrid caching strategy based on NDN-IoT [59], and distributed collaborative caching for adaptive bit rate video streaming [61]. For efficient energy consumption, work in [52] and [56] exploit Genetic Algorithm (GA) and Alternating Optimization, respectively, while to improve the total response time, authors in [42] and [62] explored Adaptive Load Balancing (ALB) and Distributed Delay-aware Caching Algorithm (DDCA), respectively. To achieve a higher cache hit rate, authors in [43] and [61] correspondingly considered Efficient Popularity-based Probabilistic Caching (EPPC), a two-stage caching scheme. Table 5 outlines the heuristic-based collaborative caching approaches, highlighting the metrics, network models, algorithms, datasets, simulation tools, results, and the limitations of each method.

Wang et al. [40] devised the NIoTE (NDN-based IoT with Edge) system by combining IoT with clustering and edge computing to make use of Named Data Networking's (NDN) advantages for more efficient data transfer. The framework uses edge devices to store data, hence lowering connection latency and costs; it also uses clustering to enable data collecting and forwarding. To control forwarding

TABLE 3. Collaborative Data Caching based on Machine Learning Models - Part 1.

Ref	Year	Metrics	Network Model	Algorithms	Dataset Used	Simulation	Result	Limitation
[31]	2021	Cache hit ratio, content acquisition delay and optimal cache placement	VECN	K-means clustering, LSTM and Q learning	Not specified	Not specified	Reduced Content acquisition delay and improved hit ratio	Does not analyze the impact of vehicle mobility on caching decision
[32]	2021	Microservice retrieval delay and cache hit ratio	MEC-enabled IoT systems	MDP , D3QN	Not specified	Python with PyTorch	Reduce average fetching delay and increase the cache hit ratio	Does not address real-world deployment challenges
[48]	2021	Task offloading and caching, total time and energy overhead	MEC	Q-learning and DQN	Not Specified	Python Numpy and TensorFlow	Model considerably reduces time and energy costs	Does not consider user mobility and data security
[64]	2021	Cache hit ratio, content retrieving delays	5G-Enabled IoT Networks	MDP, DRL-CCC, Double DQN, Dueling DQN, IBBM-RRA	Not Specified	Not Specified	Improved content cache hit rate and Reduced retrieval latency	Scalability issue in ultra-dense networks
[33]	2021	QoE, Content retrieval delay, average number of hops, cache hit ratio	ICN-IoT	AI-based collaborative filtering	Not Specified	Icarus	Cache hit rate increased by 15%, 12% and minimized retrieval delay and average hop count by 28%	Does not consider larger attributes set and scalability to larger networks
[50]	2021	Search accuracy, delay, energy consumption	CPS	TVFE, ESDC	IntelLab and Parking Birmingham	Python 3.7, TensorFlow 2.0, and MATLAB R2019a	Enhanced search accuracy, reduced search delay, and lowered energy consumption	Need to enhance search efficiency and user privacy protection further
[34]	2022	Content access delay, cache-hit ratio, device mobility, content popularity	IIoT Edge	KoMC, LSTM, DQN	The Big Data Challenge	Not Specified	Enhanced cache-hit rate and reduced content access delay for IIoT scenarios	Need more accurate content popularity prediction
[38]	2022	Service cost, service latency	IoT	UACSC, MARL, Q-Learning and CUCB algorithm.	Not Specified	Python	Achieved better overall utility, lower service cost and latency	Need to explore dynamic edge server participation and incentive schemes
[35]	2022	Cache hit rate, content access delay	VECN	RNN, MDP, Multi-agent Competitive DQN, FL	MovieLens 1M	Not Specified	Improves the cache hit rate by 19.8%, reduces content access latency by 12.5%	Content prediction accuracy degrades due to data discrepancies among user equipment

and improve request aggregation, a forwarding interest base (FIB) construction and data transmission mechanism is introduced. Experimental findings show NIoTE reduces latency and communication costs. Optimizing FIB management and request aggregation and testing the framework's scalability and IoT applications will occupy future efforts. Song et al. [41] developed a cooperative caching approach for unstable edge computing networks, addressing request routing and content placement. The Greedy Approximation Algorithm (GAA) reduces computing burden, whereas the Primal-Dual Algorithm (PDA) uses Lagrangian dual decomposition with a subgradient technique for better caching. The study shows considerable cache performance and content retrieval latency improvements over previous approaches. While less optimal than the primal-dual in some measures, the greedy approach works well in scarce resource and node settings. Future work involves adding user mobility to address local request popularity and exploring energy-efficient adaptive distributed caching techniques.

TABLE 4. Collaborative Data Caching based on Machine Learning Models - Part 2.

Ref	Year	Metrics	Network Model	Algorithms	Dataset Used	Simulation	Result	Limitation
[67]	2022	Accumulated rewards, low-latency services, Number of accepted applications	IoT	MDP, DRL, EDPN, T3DPG, WATQD	Not specified	Pytorch 1.81 environment	Superior performance in application acceptance and latency rewards	Training process requires extensive computational resources
[39]	2023	Computing power, latency, data privacy	VECN	MDP, A3C, DRL	Not Specified	TensorFlow, Python	Achieved significant advantages in system efficiency and billing data security	Investigate consensus protocols
[66]	2023	Service performance, user experience, data privacy	IoT	TFPR, secure k-NN	Kindle store, home and kitchen, and patio lawn and garden	Python and C in Ubuntu	Improved privacy preservation, reduced data service time, and enhanced performance	Need to further refine personalized secure data services for IoT systems
[51]	2023	Makespan, energy consumption, completion time.	IoT	DINM, DQN, WFS	Inspiral-30, Epigenomics-24 and Cybershake-30	ANSYS	Reduce energy consumption and improve data processing performance	Require AI-based cooperative workflow scheduling for dynamic IoT node management
[49]	2023	Cache hit rate, search cost, content popularity	AIoT	LSTM and Q-Learning	Home and Kitchen	Python 2.7 and MATLAB R2018a	Improves cache hit ratio, freshness of entity data and reduces search costs	Requires more sophisticated management of dynamic cache decisions for complex large user environments
[55]	2023	Task offloading, total system computing cost, delay	VECN	GS, G+DQN, MDP	Not specified	Python 3.7	Improved the service and computing resource utility, reduced system cost and delay	The model is computationally intensive due to a combination of DRL and Gibbs sampling, complicating real-time system implementation
[36]	2023	Cache hit ratio, communication and computation overheads	MEC	DRL-CA, lightweight cache eviction algorithm, FL	Data stream generated using Zipf distribution and MovieLens	Emulation	Outperforms other caching schemes regarding hit ratio, cache size, and communication overhead	Does not address time-sensitive caching and user mobility, which will be considered in future research
[65]	2024	Cache hit ratio, Average access delay, content popularity	MEC	GCN, GRU	MovieLens-10M	Not Specified	Better content popularity prediction, resulting in enhanced cache hit ratios and reduced average access delays	Unacceptable model training time overhead due to increasing number of nodes

A Genetic Algorithm (GA) based scheme of collaborative caching (CCS) for the Content-Centric IoT (C2IoT) is presented in [52] to optimize energy consumption and minimize redundant content transmission. The node collaboration and caching efficiency are enhanced in this approach by combining on-path and off-path caching techniques, making it a promising model for reducing network delay and energy consumption compared to existing schemes. Exploring multi-request scenarios with multiple transmission policies and blockchain integration to secure content transmission

is considered a future scope of work. Li et al. [62] proposed a four-layer IoT architecture with the collaboration of Cloud–Edge–End (CEE) that implements a hierarchical data transmission strategy and adaptive load balancing (ALB) to optimize resource allocation and improve system performance. Significant improvements are demonstrated in data transmission rates and latency minimization compared to existing frameworks. Future work to increase efficiency and performance is discussed, integrating several communication protocols, creating layer-specific data forwarding models, and optimizing device setup depending on data quantities.

Further, to reduce response times for IoT users, Kim et al. [42] enhanced the caching of program data and replicas across several cloudlets. By means of simulations, they have established the efficiency of their heuristic technique, Distributed Delay-aware Caching Algorithm (DDCA). Significant improvements are demonstrated in response time compared to existing methods; however, the focus should be on refining the model for diverse application types and exploring predictive caching strategies. Fan et al. [56] developed a technique of resource management for WiFi-cellular heterogeneous networks with MEC that combines job offloading and service caching. The study proposed an effective solution to maximize energy benefits and minimize delay by considering multiple factors such as the MEC servers' processing and storage capabilities and the distribution of cellular channels. The alternating optimization decomposed into 3 subproblems: the branch-and-bound algorithm (BBA) for task offloading, variable relaxation with penalty functions (VRPF) for service caching, and the difference of convex and Kuhn-Munkres algorithm (DCKMA) for channel allocation. Simulation work demonstrates energy consumption is reduced significantly compared to existing approaches.

Naeem et al. [59] proposed a hybrid caching approach based on named data networking (NDN), termed NDN-IoT. This method enhances network performance by strategically organizing material according to request frequency and timestamps. It utilizes efficient probabilistic caching based on popularity (EPPC), incorporating effective eviction policies, content selection, and strategic caching at the edge and primary nodes. The simulation results illustrate the higher route stretch ratio, reduced latency, and enhanced cache hit rates. This adaptable approach enhances IoT networks through potential extra node allocation and optimization of caching strategies. Li et al. [61] proposed a technique for adaptive bit-rate video streaming that uses networked collaborative caching and processing. A two-stage caching scheme is introduced that uses the popularity of the content over the long and short terms to optimize cache placement using the Proactive Cache Placement Algorithm (PCPA) and updates the cache by employing the Reactive Cache Update Algorithm (RCUA). A real-time cooperative edge caching-based request scheduling algorithm (RRSA) is implemented. The approach improves cache hit rates, lowers backhaul connection burden, and effectively decreases initial buffer delay, enhancing the user experience for video streaming services. An adaptive system for dynamically sharing computing power and scheduling tasks (ADCS) in edge environments is suggested in [43]. ADCS employs the Greedy Decision Method (GDM) and the Best-Fit Algorithm (BFA) to successfully reduce deadline-missing rates and average task completion times. The proposed mechanism shows considerable improvement compared with existing algorithms and sets the foundation for future enhancements in microservice dependency management and open-source platform integration.

The studies examined in the heuristic-based collaborative caching category show a broad spectrum of methods targeted to maximize resource usage in IoT and MEC environments, lower latency, and improve the efficiency of cache. By combining smart algorithms with cooperative frameworks, these studies generally aim to solve the dynamic aspects of IoT environments. For instance, the use of IoT clusters with edge device caching [40] and probabilistic caching methodology employing NDN-IoT [59] underlines the efficiency of merging NDN with edge computing to attain lower latency and reduced communication costs. Collaborative caching schemes in content-centric IoT have been developed to reduce energy consumption and redundant data transmission [52], which in turn decreases network delay and improves energy efficiency. Optimization algorithms play an essential role in these advancements.

Hierarchical systems employing adaptive load balancing strategies [62] have demonstrated notable increases in data transmission speeds and latency reductions. Heuristic approaches for cache program data and replication over multiple cloudlets have helped Internet of Things users maximize response times [42]. A resource management technique that integrates job offloading and service caching [56] to optimize energy efficiency and reduce delay is proposed, taking into account processing and storage capabilities as well as the distribution of cellular channels. Adaptive systems for the dynamic sharing of computing power and task scheduling [43] in edge environments have been proposed to decrease rates of missed deadlines and to optimize average task completion times.

In summary, these research emphasize the need to include intelligent algorithms, cooperative frameworks, and adaptive techniques in designing efficient caching mechanisms for edge computing in Internet of Things environments. Future studies should concentrate on scalable systems, address dynamic network conditions, energy efficiency, and guarantee data security and privacy to fully realize the potential of edge computing in IoT applications.

D. COLLABORATIVE DATA CACHING BASED ON HYBRID MODEL

By integrating machine learning, stochastic, and heuristic techniques, a combination of methods allows for utilization of the strengths of each particular model. Such a hybrid strategy is often utilized in collaborative caching

TABLE 5. Heuristic Model-based Collaborative Data Caching.

Ref	Year	Metrics	Network Model	Algorithms	Dataset Used	Simulation	Result	Limitation
[40]	2021	Communication latency and transmission cost	NDN-IoT	FIB and Data Communication Algorithm	Not specified	Not specified	Reduces latency and costs by optimizing request aggregation	FIB explosion and low request aggregation may occur in large-scale IoT environments
[41]	2021	Content access delay, cache hit rate resource utilization,	CDN	PDA, GAA	Not specified	Mathematical model (Watts-Strogatz)	Model outperforms in hit rate and content retrieval delay	Impact of user mobility is not considered
[52]	2022	Energy consumption and redundant content transmission, QoE	C2IOT	CCS, GA, Zipf's law	Not Specified	Not Specified	Reduced network delay, energy consumption, and efficient resource utilization	Required precise content popularity prediction
[62]	2022	QoS, response time, operating costs	IoT	ALB	Not specified	MATLAB R2020b	Improves resource utilization and packet delivery rates	Challenges in maintaining adaptive load balancing across diverse network conditions
[42]	2022	Total response time	IoT	DDCA	Not specified	Not specified	Achieves a lower response time compared to baseline	Does not consider diverse IoT scenarios
[56]	2022	Energy efficiency, task delay	MEC	BBA, VRPF, DCKMA	Not specified	Not specified	Achieves greater energy reduction compared to other schemes	Tested in WiFi networks, need testing in more heterogeneous environments
[59]	2022	data dissemination, latency, cache hit ratios	NDN-IoT	EPPC	Not specified	Icarus	Improves cache hit ratio, reduces average latency, and path stretch	The strategy may not perform well with highly volatile popularity
[61]	2022	Average initial buffer delay, user experience, cache hit rate	MEC	PCPA, RCUA, RRSA	Not specified	Matlab	Efficiently reduces initial buffer delay, backhaul link load, improves cache hit rate	Depends heavily on accurate long and short-term video popularity predictions
[43]	2023	Deadline missing rate, average task completion time	MEC	GDM, BFA	Google Cluster 2019	Java8	Reduces deadline missing rate and average task significantly	Does not account for data and logical dependencies between microservices

to enhance performance by combining the efficiency of heuristic approaches, the robustness of probabilistic decision-making, and the accuracy of machine learning. Researchers integrate these models to optimize resource utilization, enhance cache placement strategies, and reduce latency in edge computing. Here, we present a few of the referenced works on collaborative content caching using a combination of models.

Various innovative ideas and methods have been proposed to address the challenging issue of collaborative caching in IoT and edge computing systems. Several often investigated metrics for collaborative content caching have been considered in the articles referred to based on the hybrid caching model. A few of the most often investigated metrics that significantly affect caching performance include latency [44], [45], [60], [63], cache hit ratio [45], [60], resource utilization [60], [68], energy consumption [57], QoS [44], prediction accuracy, load balancing, and offloaded traffic [63]. The authors have investigated the proactive caching with prefetching scheme [60], heuristic collaborative caching strategy [44], adaptive caching algorithms [45] to minimize communication latency and content acquisition

delay. To increase the cache hit rate, authors have taken into consideration a three-fold solution including delay-aware replacement, association-based caching, and the MEC collaborative caching Algorithm [60] and transcoding-enabled caching [45], whereas to reduce bandwidth consumption and to optimize resource utilization, the collaborative greedy algorithm (CGA) [60] is explored. To optimize energy consumption a deep neural network-based optimization module [57] has been employed. On the other hand, to improve QoS a heuristic strategy with k-set packing is adopted in [44] and to optimize prediction accuracy, load balancing, and offloaded traffic, a multi-objective evolutionary algorithm based on knowledge mining (KMaOEA) is explored in [63]. The hybrid model-based collaborative caching is summarized in Table 6, summarizing the metrics, network models, algorithms, datasets, simulation tools, results, and the limitations of each method.

Ugwuanyi et al. [60] proposed a Predictive Collaborative Replacement (PCR) caching scheme for MEC that combines an efficient replacement algorithm, the Least Frequently Used Selective Historical method (SHLRFU), an Algorithm for Proactive Prefetch Caching (PPCA), Collaborative Greedy Algorithm (CGA). By considering the factor of cold cache objects, adapting dynamically to user request patterns, and enabling effective cache sharing among MECs, the PCR system successfully lowers latency and increases cache hit ratios. Future research proposes modifying the algorithm for ICN scenarios and investigating decentralized learning mechanisms. Chen et al. [57] developed a collaborative entity state data caching technique between the edge and the cloud to improve the performance of MEC systems. By optimizing cache placement, computation resources, and bandwidth allocation based on user preferences and locations, the system remarkably minimizes energy consumption and latency. The mixed-integer non-convex issue is successfully handled by a deep learning-based Discrete Action Generation employing a Stochastic Quantization (SQDAG) technique, attaining almost optimum performance. Future studies will concentrate on enhancing channel model accuracy, generalizing task correlation, and dynamic cache decision-making. Zeng et al. [44] considered edge-cloud collaboration to minimize the latency associated with data support for VECN. The proposed SDN-based framework splits the VECN coverage into dense and ordinary zones, clusters edge servers using an enhanced K-mean++ algorithm, and partitions server storage into local, group, and global sections for optimal data caching. The problem is modeled as an NP-hard weighted k-set packing problem (WkPP), and to minimize data migration delays and improve QoS, a heuristic strategy is developed. Results demonstrate that the suggested approach performs better than alternative caching solutions in terms of lowering latency and enhancing application performance; nevertheless, its adaptability in highly dynamic environments could be a limitation.

Xiao et al. [45] devised an online caching framework facilitated with transcoding (ToC) for IoT video services, addressing the challenges of real-time decision-making in heterogeneous environments. The task of caching is modeled as a problem of Online Convex Optimization (OCO), and solutions are developed for both stable and elastic IoT networks. A dual update policy that uses Partial Lagrangian Functions (PLF) is introduced to handle time-varying resources while maintaining performance within resource constraints. The study demonstrates significant latency reduction and QoE enhancement improvements by integrating computation into caching mechanisms and developing adaptive algorithms for stable and elastic networks. The higher performance of the proposed method in many scenarios is confirmed by theoretical arguments and comprehensive analysis using several datasets. The popularity prediction approach for cooperative caching based on many-objective optimization and a framework that optimizes different network metrics for collaborative IoT networks spanning cloud, edge, and end was investigated in the paper by Hu et al. [63]. It utilizes a number of objective-based knowledge mining evolutionary algorithms (KMaOEA) for knowledge mining optimization, combines three prediction approaches for content popularity estimation, and includes horizontal and vertical cooperative caching methods. Simulations demonstrate significant improvements in performance. Future work will focus on incorporating intelligent routing strategies to further enhance the framework's capabilities.

The research on hybrid model-based collaborative caching has presented novel methodologies for data caching in edge computing, including online caching frameworks with transcoding functionalities [45], collaborative entity state data caching between edge and cloud [57], and predictive collaborative replacement caching strategies [60]. Additionally, SDN frameworks for VECN [44] seek to increase QoS and minimize data transfer delays. While these contributions are valuable, their adaptability in highly dynamic environments remains a challenge. A thorough study reveals that much of the work focuses on specific technical solutions without fully addressing the broader challenges inherent in dynamic and heterogeneous edge environments.

Development of adaptive, scalable, context-aware caching systems that can dynamically react to changing network conditions and user demands is necessary if we are to progress the area. Although the body of present research provides a platform for data caching in edge computing, there is still a tremendous possibility for innovation employing extensive, system-level solutions addressing the complexity of modern network environments.

E. ASSESSMENT AND ANALYSIS

This section explores and summarizes the discussion on the review of collaborative data caching in edge computing, presenting key aspects like classification of collaborative data caching models, type of network employed, simulation tools and dataset used, and key performance metrics considered in articles on collaborative data caching in edge computing. We have divided collaborative data caching in edge

TABLE 6. Collaborative data caching based on hybrid model.

Ref	Year	Metrics	Network Model	Algorithms	Dataset Used	Simulation	Result	Limitation
[60]	2021	Cache hit ratio, access delay, and resource utilization	MEC	SHLRFU, PPCA, CGA	MovieLens 20M	GNS3	Improved cache hit ratio and reduced access delay	Employs a centralized learning model
[57]	2022	Energy consumption, service latency,	MEC	DNN, SQDAG	Not Specified	MATLAB R2020a	Saved 33% and 6.69% of energy consumption compared to baseline methods	A constant cache decision size may lead to increased computation overhead
[44]	2023	Acquisition delay, QoS	VEC	K-mean++, WkPP	Not specified	Not specified	Reduce delay by 50% and 30%, improve the QoS by 40% and 20% compared to the LRU, LFU and Random strategies	May face challenges in highly dynamic, real-world VEC scenarios
[45]	2024	Delay, Regret, QoE, Cache hit rate, Constraint violation	IoT	OCO, PLF, DR	LDOS-CoMoDa, MovieLens, Movie Tweet, Yahoo	Python	Improved learning ability, cache hit rate, and lower latency in dynamic IoT settings	Performance might be less optimal in unpredictable environments
[63]	2024	Optimize prediction accuracy, delay, offloaded traffic, and load balancing	IoT	KMaOEA	MovieLens 100K	Not Specified	Improves data caching efficiency through improved prediction accuracy	High computational complexity associated with many-objective optimization

computing into four different categories: model-based, which includes stochastic, game theory, and mathematical models; machine learning-based, which uses AI for prediction and optimization; heuristic, which provides lightweight cache management algorithms; and hybrid models, which combine several strategies for increased adaptability. Each of these models addresses distinct issues with cache placement, replacement, and coordination. Figure 4 shows the percentage of various types of models employed for collaborative data caching. ML-based models with 17 studies are the most widely adopted methods by researchers for collaborative data caching in edge computing.

Collaborative data caching techniques are evaluated across various network architectures, including MEC, IoT, and vehicular edge networks. The variety of environments highlights the necessity for flexible caching solutions. Figure 5 shows the statistical details of various types of networks employed by researchers for collaborative content caching in the edge computing environment. Figure 6 and figure 7 show datasets and simulation tools employed by researchers in different studies. 18% of research used Matlab as a platform, while 26% of studies used Python. However, many researchers did not explicitly mention the details of the dataset and simulation tools used for implementing and testing the model. Figure 8 shows the metrics considered for a performance evaluation of the collaborative caching model in edge computing. Most researchers have considered delay and

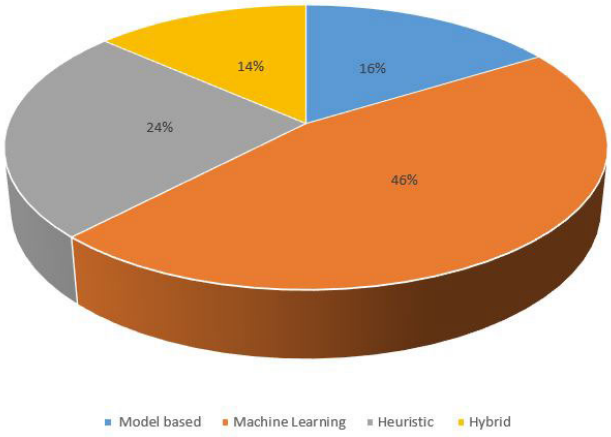


FIGURE 4. Percentage of various approaches for collaborative data caching.

cache hit rate metrics to evaluate their model's performance. It is important to note that many articles that were examined had considered multiple objectives for evaluation.

VII. OPEN CHALLENGES

The primary focus of collaborative data caching in IoT-based edge computing is to manage the complexity, scalability, and dynamic nature of today's ever-growing modern network.

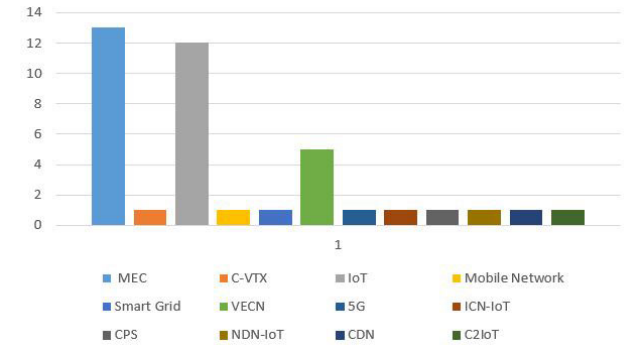


FIGURE 5. Types of network employed for Collaborative caching in Edge computing.

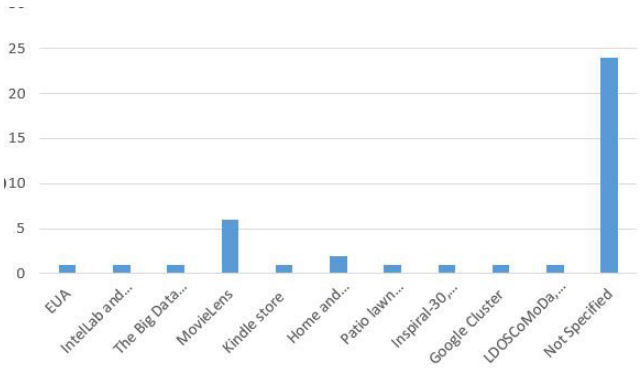


FIGURE 6. Dataset utilized in collaborative edge caching.

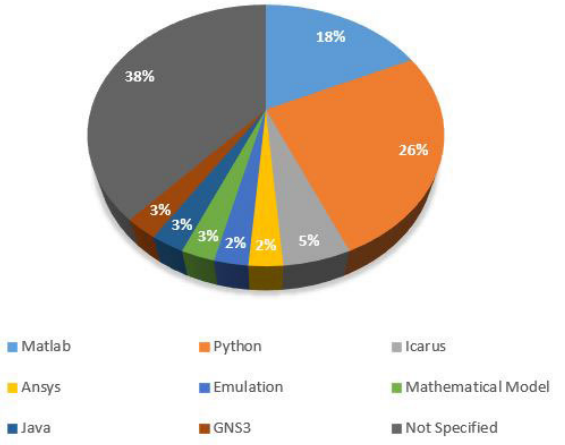


FIGURE 7. Simulation tools employed in collaborative edge caching.

This section explores several open challenges and issues associated with collaborative data caching in edge computing.

A. SCALABILITY IN LARGE-SCALE HETEROGENEOUS NETWORKS ENVIRONMENTS

The rapid expansion of heterogeneous IoT networks consisting of several technologies, such as WiFi, cellular networks, and IoT-specific protocols, provides a significant challenge to the development of scalable caching systems. These networks must maintain high performance while supporting various characteristics, such as dependability, latency, and

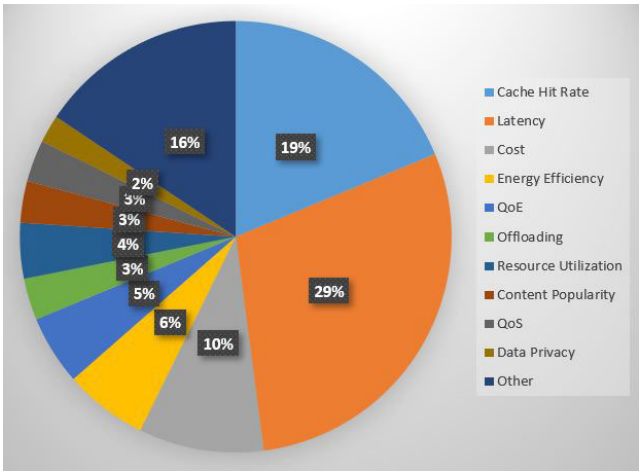


FIGURE 8. Metrics adopted for performance evaluation in Collaborative edge caching.

bandwidth, across many network technologies. The design of adaptive caching solutions for heterogeneous networks requires sophisticated approaches that dynamically react to changes in network capacity, data load patterns, and the several capabilities and restrictions of different network technologies. Furthermore, load balancing methods and better resource allocation algorithms are needed to manage expanding data volumes across different networks, thereby optimizing performance across several network kinds. Effective caching systems have to be scalable across different environments, handling data across several network protocols and architectures and maintaining system performance. The solution calls for fresh approaches that might fit the unique characteristics of every kind of network and guarantee seamless integration and efficient operation across the entire system.

B. SECURITY AND PRIVACY CONCERNS

Concerns regarding security and privacy in collaborative edge caching are significant issues that have garnered considerable attention in recent times. As the data processing and storage occur at the network's edge, close to end users. The sensitive characteristics of IoT data, coupled with the requirements of edge computing, necessitate the implementation of robust security and privacy solutions that maintain system efficiency while safeguarding data. Recent research findings demonstrate effective approaches to these challenges, enhancing privacy protection and reducing service time. Further research is necessary to establish a framework for secure data storage that maintains efficient caching and retrieval mechanisms while minimizing the risks of unauthorized access, data breaches, and illegal modifications. This framework should integrate advanced encryption techniques, secure multi-party computation methods, and other AI-driven solutions, ensuring local data privacy protection among edge servers. Future research should concentrate on developing advanced strategies for collaborative caching that maintain privacy, as well as

employing quantum-secure protocols to mitigate emerging security risks while ensuring the efficiency of caching mechanisms.

C. CONTEXT-AWARE CACHING

Context-aware caching signifies a caching strategy in which individual caches or cache policies adjust to the changing environment of the applications, devices, or users. By examining the location, mobility patterns, machine attributes, and even the current network, this method would improve cache placement. Such analysis helps the system to predict the demand for the content and cache it accordingly in the vicinity of the user, thus lowering latency, saving energy, and optimizing bandwidth utilization. It will also be essential to develop adaptive algorithms to verify the context on the ground in realtime to improve the efficiency and performance of cache storage systems in the IoT environment.

D. ENERGY-EFFICIENT CACHING

Energy-efficient caching techniques focus on minimizing the energy consumption for performing the caching processes, which is vital due to the energy constraints of edge nodes and IoT devices. Conserving energy while optimizing the cached content placement and retrieval is the key element for the sustainability of the system. Further research is required to develop mechanisms that effectively balance energy consumption with cache optimization and performance metrics. These algorithms might include dynamic changes in the cache placement according to the state of the environment, energy level, users' activity, and general network conditions. The energy-efficient caching mechanism development will be a crucial step towards ensuring the sustainability and efficiency of organizations achieving large-scale operations within IoT networks.

E. DYNAMIC CONTENT POPULARITY PREDICTION

Content popularity changes over time and space. Traditional static caching strategies are inefficient due to the dynamic nature of data. Hence, dynamic content popularity prediction becomes an essential part of collaborative data caching. It involves forecasting variations in user preference over time. Researchers should focus on the creation of complex AIML-based algorithms for real-time prediction of content popularity for proactive cache placement and upgrades. To overcome the remarkable challenges presented by dynamic IoT ecosystems, the ultimate goal is to enable effective and efficient content distribution while taking into account the constantly changing needs of users.

F. CACHE CONSISTENCY MANAGEMENT

In a collaborative edge environment characterized by multiple IoT devices, diverse data formats, and distinct edge servers for data storage and analysis, data inconsistency may arise. This is particularly evident when devices work together on tasks that require shared or interdependent data. Maintaining consistency across multiple cache sites is essential to prevent end users from accessing outdated data. Future research may concentrate on effective methodologies for version control systems, consistency levels and protocols, and real-time cache updates to reduce synchronization latency and ensure precise and prompt data transmission. Achieving these objectives without substantially increasing network overhead or degrading system performance presents a considerable challenge in this field.

G. MOBILITY MANAGEMENT

Mobility management is a crucial consideration and an important area of study for collaborative caching in edge computing, particularly in mobile users and IoT environments. As users and devices move across the different network segments, providing uninterrupted access to the cached content becomes difficult. The main goal of the research is to design mobility-aware caching methods that can dynamically change to fit the user's movement patterns. These techniques should provide low-latency services and lower the number of cache misses in mobile environments. Attention may also be diverted in the future towards incorporating prediction models that aim to understand user behaviors to carry out proactive caching of the necessary data. Moreover, research efforts may also focus on creating effective handover mechanisms that ensure continuous service delivery across various network domains or edge nodes.

H. INTEGRATION WITH EMERGING TECHNOLOGIES

Integrating AI and ML into caching strategies aims to improve their intelligence and adaptability by utilizing algorithms to study user behavior, content popularity, and network states. These sophisticated computational approaches might then maximize real-time cache eviction choices and cache placement, hence producing a more efficient, flexible, and elastic caching system. Furthermore, it is feasible to automate resource allocation, future content demand prediction, and gradually enhance cache utilization. AI-based caching might produce better latency and energy economy, hence producing an efficient and scalable caching system fit for any other IoT application.

I. EDGE FOG CLOUD COLLABORATION

Collaboration in edge, fog, and cloud computing efficiently assigns caching activities, respectively, to achieve the right balance among the desired latency, storage, and computing power of the system. This approach can enhance system performance and resource utilization by making use of the immediate vicinity of the edge and fog nodes for low-latency services and pushing complex jobs to the cloud. Research could concentrate on collaboration protocols, effective data migration, and dynamic allocation methods to improve service delivery in various IoT contexts.

VIII. CONCLUSION

In edge computing, collaborative data caching provides a feasible way to cope with the massive data needs of contemporary IoT and data-intensive applications. Edge

caching makes it possible to reduce latency and bandwidth usage and enhance scalability by moving data processing and storage closer to end users. These advantages are crucial for enabling time-sensitive and bandwidth-intensive services. In this paper, we have presented a survey of several articles on collaborative data caching in an edge computing environment. This review divided caching strategies into four categories: heuristic, mathematical, and game theory; machine learning-based; stochastic; and hybrid models. Each model has special advantages for handling data in a range of network scenarios. Despite major progress, important problems continue to exist, including managing dynamic network environments, guaranteeing energy efficiency, and preserving data consistency. Addressing these open issues will be crucial for maximizing edge caching efficiency and enabling the full potential of edge computing to support upcoming IoT-driven applications.

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